

On wave propagation in solids and altermagnets

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Guide wave signals in solids

One can use different excitations: electrons, photons, phonons, etc.

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1 Unidirectional

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1 Unidirectional

2 Robust

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One can use different excitations: electrons, photons, phonons, etc.

- 1 Unidirectional
- 2 Robust $\longrightarrow$ topologically protected

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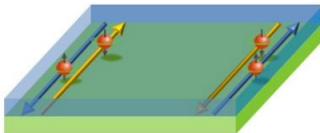
Guide wave signals in solids

One can use different excitations: electrons, photons, phonons, etc.

- 1 Unidirectional
- 2 Robust $\longrightarrow$ topologically protected
- 3 Spin-momentum locking

Examples

Example 1: quantum spin Hall effect



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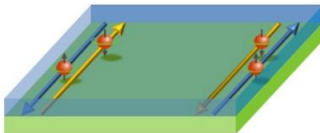
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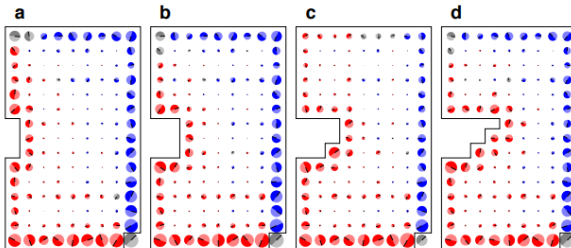
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Examples

Example 1: quantum spin Hall effect



Example 2: 2D array of pendula



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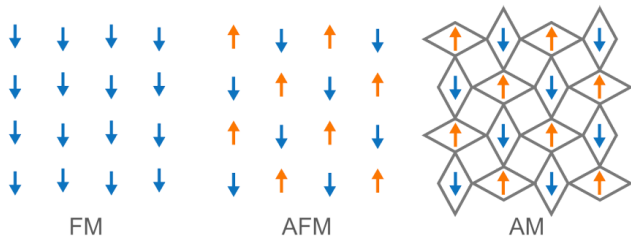
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Kramer's theorem: $\text{TRS} + \text{half-integer spin} \implies \text{degenerate energies}$

What are altermagnets? A different phase of matter II

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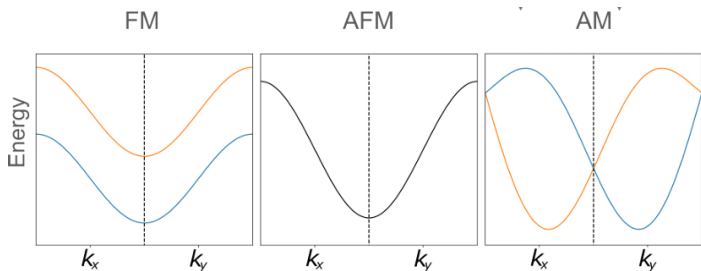
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- 1 Symmetry: Rotation+TR denoted by $C_4\mathcal{T}$
- 2 Zero magnetization
- 3 Spin-split bands
- 4 $C_4\mathcal{T} \implies N_{\uparrow} = N_{\downarrow}$



Altermagnetic Lieb lattice

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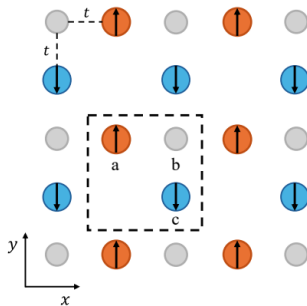
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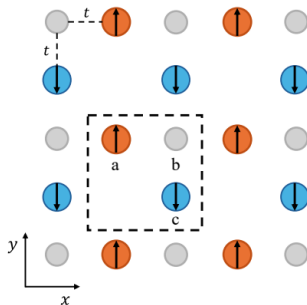
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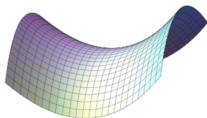
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It has hyperbolic bands:



Altermagnets have interesting properties

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1 Anisotropic wave propagation

- Unidirectional: control wave propagation direction
- Spin-momentum locking: different spins propagate in different directions

2 Piezomagnetism: magnetic moment is induced by strain

Useful in spintronics, spin-splitters, ultrafast photomagnetism, thermoelectrics ...

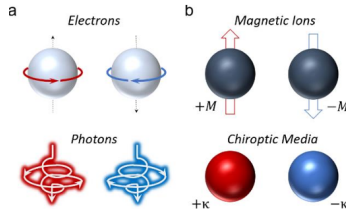
Generalization of alter symmetry

Generalization alter symmetry

The spin can be substituted by any other 2-state variables.

Some examples are:

- Photon polarization $\rightarrow$ photonic altermagnet
- Electric dipole $\rightarrow$ alterelectric
- Pendula chirality $\rightarrow$ mechanical altermagnet (metamaterial)



Example 1: piezoelectricity

- Onsager relations:

$$\varepsilon_{ij} = S_{ijkl}^E \sigma_{kl} + d_{ijk} E_k,$$

$$D_i = d_{ijk} \sigma_{jk} + \kappa_{ik}^\sigma E_k,$$

- Restrictions due to crystal symmetry:

$$[d_{ij}]^T = \begin{bmatrix} 0 & 0 & 0 & d_{15} & 0 \\ 0 & 0 & 0 & d_{24} & 0 \\ d_{31} & d_{32} & d_{33} & 0 & 0 \end{bmatrix}^T \quad (j = 1, 2, 3, 4, 5, 6)$$

- Enhancement due to complex structures
- Properties given by its structure, not by its parts

Metamaterials II

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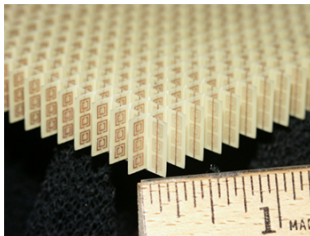
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Example 2: negative refractive index



Non-hermiticity

■ Motivation: open systems

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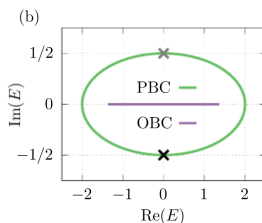
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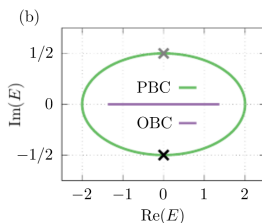
Non-hermiticity

- Motivation: open systems
- Spectrum becomes complex



Non-hermiticity

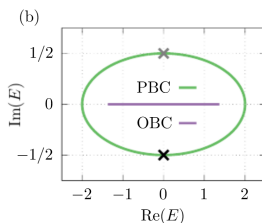
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- Two major effects: bulk-boundary correspondence and non-hermitian skin effect

Non-hermiticity

- Motivation: open systems
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- Two major effects: bulk-boundary correspondence and non-hermitian skin effect
- Example: Hatano-Nelson model

Non-hermiticity

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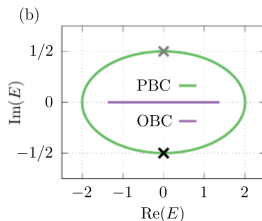
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- Motivation: open systems
- Spectrum becomes complex



- Two major effects: bulk-boundary correspondence and non-hermitian skin effect
- Example: Hatano-Nelson model

$$\hat{H}_{\text{HN}} = \sum_j \left[t_L c_{j+1}^\dagger c_j + t_R c_j^\dagger c_{j+1} \right]$$

Non-reciprocity

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- Motivation: classical analogue of non-hermiticity

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- Motivation: classical analogue of non-hermiticity
- Breaks inversion symmetry $\rightarrow$ unidirectional waves

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- Response differs when transmitter/receiver swapped

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- Motivation: classical analogue of non-hermiticity
- Breaks inversion symmetry $\rightarrow$ unidirectional waves
- Response differs when transmitter/receiver swapped
- Breaks Newton's third law (classical level)

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- $$\begin{pmatrix} F_1 \\ F_2 \end{pmatrix} = \begin{pmatrix} k & -k^o \\ +k^o & k \end{pmatrix} \begin{pmatrix} \Delta_1 \\ \Delta_2 \end{pmatrix}$$

Non-reciprocity

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- Non-reciprocal units

Non-reciprocity

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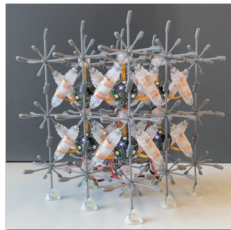
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- Non-reciprocal units
- Leads to pendula chirality



In practice

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The procedure I have been following is:

- 1 Choose a classical model

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The procedure I have been following is:

- 1 Choose a classical model
- 2 Write equations of motion

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The procedure I have been following is:

- 1 Choose a classical model
- 2 Write equations of motion
- 3 Multiple scale expansion

In practice

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The procedure I have been following is:

- 1 Choose a classical model
- 2 Write equations of motion
- 3 Multiple scale expansion
- 4 Map to quantum system

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The procedure I have been following is:

- 1 Choose a classical model
- 2 Write equations of motion
- 3 Multiple scale expansion
- 4 Map to quantum system
- 5 Analyze spectrum and eigenstates

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- 5 Analyze spectrum and eigenstates

The idea is to apply them to increasingly more complex models.

Experimentally: build a Lieb lattice of non-reciprocal pendula and compare with theoretical predictions

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Goal

Study altermagnetism and non-hermiticity in classical analogues

- Altermagnets: $C_4\mathcal{T}$ symmetry can lead to unidirectional and spin-momentum locking propagation
- Non-hermitian skin effect: waves localized at boundaries
- Classical analogues enable experimental realization

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Activity

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The classical analogue of non-hermiticity of Hatano-Nelson is non-reciprocity.

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Activity: internal energy source.

Activity

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Activity: internal energy source.

There are different ways to achieve activity:

Activity

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Activity: internal energy source.

There are different ways to achieve activity:

- Negative damping

Activity

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- **Non-reciprocity**